2025 32nd Asia-Pacific Software Engineering Conference (APSEC) | 979-8-3315-6653-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/APSEC66846.2025.00055

SVPrompt-MR: An LLM-Based Metamorphic Relation Identification Method with Self-Verification Mechanism

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***Abstract*—Metamorphic testing is an effective technique to alleviate the oracle problem in software testing, with metamorphic relations (MRs) as its core. Currently, the identification of MRs relies mainly on domain experts, which leads to low efficiency and considerable subjectivity in MR formulation. Large Language Models (LLMs) show great potential for identifying MRs automatically. In this paper, we propose a self-verification-based prompt construction strategy for MR identification, which consists of two stages: MR identification and MR verification. Our method is applied to the thermal-hydraulic safety design program for pebble-bed high-temperature gas-cooled reactors, with deepseek-R1 as the underlying model. Experimental results demonstrate that our approach not only covers all 44 MRs identified by experts, but also discovers 24 new MRs. It significantly outperforms existing methods in terms of coverage rate, accuracy rate, and innovation rate, thereby exhibiting superior identification capability. The research results not only advance the study of MR identification, but also provide a valuable reference and inspiration for researchers.**

***Index Terms*—Large Language Models, Metamorphic Relation Identification, Prompt Engineering**

I. Introduction

The thermal-hydraulic safety design program for pebble-bed high-temperature gas-cooled reactors involves complex physical processes and numerous parameters, which are affected by various factors such as physical models, boundary conditions, and material properties. This complexity makes it difficult to obtain expected results, rendering traditional testing methods ineffective and leading to the well-known oracle problem in software testing [1] [2] [3].

To address the oracle problem, a variety of techniques have been proposed, including assertions [4], model checking [5], and metamorphic testing (MT) [6] [7] [8] [9] [10]. MT is widely recognized as one of the most promising approaches. It verifies program correctness through metamorphic relations (MRs) [11], without the need to construct expected outputs. Existing MR identification methods can be broadly categorized into static and dynamic approaches. Static methods primarily construct MRs through program analysis and domain knowledge inference, as exemplified by METRIC [12] and METRIC+ [13]. Dynamic methods, on the other hand, identify MRs using machine learning techniques [14] [15] and search-based software engineering [16], such as AutoMR [17], SPSS [18], and semi-supervised machine learning [14]. However, most of these methods still rely heavily on domain experts, resulting in low efficiency and considerable subjectivity in MR formulation. Large Language Models (LLMs) [19] have demonstrated exceptional capabilities in natural language processing and knowledge understanding. By learning from massive datasets, LLMs possess "world knowledge," along with strong semantic understanding and reasoning abilities, providing broad potential for MR identification. For example, Tsigkanos et al. [20] used LLMs to extract variables required for MT from software user manuals. Shin et al. [21]leveraged LLMs to automatically derive executable MRs from requirement reports and guided LLMs in metamorphic testing through few-shot prompting. Zhang et al. [22]used ChatGPT to automatically generate MRs for autonomous driving systems (ADSs). Jiaming Zhang et al. [23] evaluated the MR identification capabilities of LLMs across 37 programs. Other studies [24] have shown that GPT-4 can identify MRs of certain reference value, though manual review is still necessary to ensure their accuracy and applicability. These results indicate that LLMs can effectively generate MRs and reduce manual effort. However, these studies are still at the experimental stage, and there has been no prior work on applying LLMs to MR identification for complex programs.

For the thermal-hydraulic safety design program of pebble-bed high-temperature gas-cooled reactors, a set of MRs has already been identified by domain experts. Therefore, this study investigates LLM-based MR identification methods for complex programs. The main contributions of this paper are as follows:

- *Prompt construction strategy.* We propose a self-verification-based prompt construction strategy for MR identification, including both MR identification and MR verification stages.
- *Effectiveness of MR identification.* Our method effectively identifies MRs in complex programs, not only covering all 44 MRs recognized by experts, but also discovering

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24 new MRs. It significantly outperforms existing approaches in terms of coverage, accuracy, and innovation rate.
- *Analysis of syntactic structures.* We demonstrate which syntactic structures in software development documents contribute most to MR extraction

The remainder of this paper is organized as follows. Section II presents background knowledge on metamorphic testing and large language models. Section III reviews related work. Section IV details the research, including research assumptions, methods, and examples. Section V introduces the experimental setup and research questions. Section VI provides an analysis and discussion of the experimental results. Finally, Section VII summarizes the paper and outlines future directions.

II. BACKGROUND

A. Metamorphic Relations

In software testing scenarios where test oracles are difficult to obtain, metamorphic testing (MT) has become an invaluable testing approach [25]. The core objective of MT is to detect faults by leveraging the inherent properties of the system under test (SUT). These essential properties are embodied as objective and necessary correlations between multiple inputs and their corresponding expected outputs—these correlations are defined as *metamorphic relations* (MRs). Clearly, a set of MRs forms the central component of MT. A representative type of MR can be defined as follows:

Given a program P that implements the function $f(x)$, for n inputs $(x_1, x_2, \ldots, x_n)$, each input yields an output $\{f(x_1), f(x_2), \ldots, f(x_n)\}$. If there exists a relation r among the inputs and a relation R among the outputs, and whenever $r(x_1, x_2, \ldots, x_n)$ holds, $R\{f(x_1), f(x_2), \ldots, f(x_n)\}$ must also hold, then (r, R) is called a metamorphic relation of P, where r is the input relation and R is the output relation.

The development of scientific computing programs typically follows multiple stages, including physical modeling, algorithm design, and program implementation. Domain knowledge at each stage provides support for MR identification. A hierarchical model for classifying metamorphic relations has been previously proposed [26], as shown in Fig.1, which systematically guides the sources and propagation of MRs. This model divides MRs into four levels, corresponding to mathematical physical equations, numerical algorithms, program specifications, and code execution traces.

Most static methods focus on the physical model and computational model, while dynamic methods mainly target the code model and execution trace. Compared with code models and execution traces, physical models and computational models are richer in semantic information. Their corresponding MRs can more profoundly reflect the complex physical mechanisms, mathematical principles, and inherent constraints upon which the program relies.

B. Large Language Models

Large Language Models (LLMs) have demonstrated powerful performance across a variety of tasks and have been widely applied in areas such as text generation [27], code completion [28], and program repair [29]. These models are pretrained on massive amounts of data from the internet, endowing them with the capability to understand and generate natural language. In practical applications, users can guide the model to accomplish specific tasks simply by providing natural language prompts, without the need for additional training or specialized data. This capability makes prompt design particularly crucial and has given rise to the concept of "prompt engineering"—the practice of constructing effective prompts to enhance task performance. Existing studies have shown [30]that combining natural language descriptions with a few example inputs can significantly improve LLM performance on specific tasks.

LLMs can be classified into three categories: decoder-only models, encoder-only models, and encoder-decoder models. Recent research indicates that decoder-only models achieve superior results on MR identification tasks. DeepSeek-R1 [31]adopts a decoder architecture and integrates a Mixture of Experts (MoE) mechanism. It innovatively employs reward model-driven training, which substantially improves its performance on reasoning, mathematical, and various other benchmark tasks. Given that the thermal-hydraulic safety design program for pebble-bed high-temperature gas-cooled reactors involves a large number of complex physical processes, the identification of MRs demands strong reasoning and comprehension abilities. Therefore, this paper selects DeepSeek-R1 as the base model.

III. RELATED WORK

In recent years, numerous studies have sought to enhance the effectiveness and efficiency of metamorphic relation (MR) identification from various perspectives. The main approaches can be categorized into traditional MR identification methods and LLM-based identification methods.

A. Traditional MR Identification Methods

Early MR identification primarily relied on manual experience, expert knowledge, and rule-driven approaches, such as manual analysis, static code analysis, and relation induction based on test cases. Chen et al. proposed METRIC [12], a classification-selection framework, and further extended it to METRIC+ [13] by incorporating output domain processing. With the increasing demand for automation, the field has seen a variety of explorations based on NLP and program analysis. Blasi et al. introduced the MeMo [32] framework, which uses NLP techniques to automatically identify equivalence MRs from Javadoc comments and generate executable assertions, advancing the automation of MR identification. However, its effectiveness heavily depends on the quality of comments and declines significantly with non-standard documentation. Xu et al. proposed MR-Scout [33], which mines MRs from test cases to improve test completeness, but its applicability is limited by the need for high-quality test suites. Bo Zhang et al. presented the AutoMR [17] method, capable of automatically generating high-quality, diverse MRs. Kanewala

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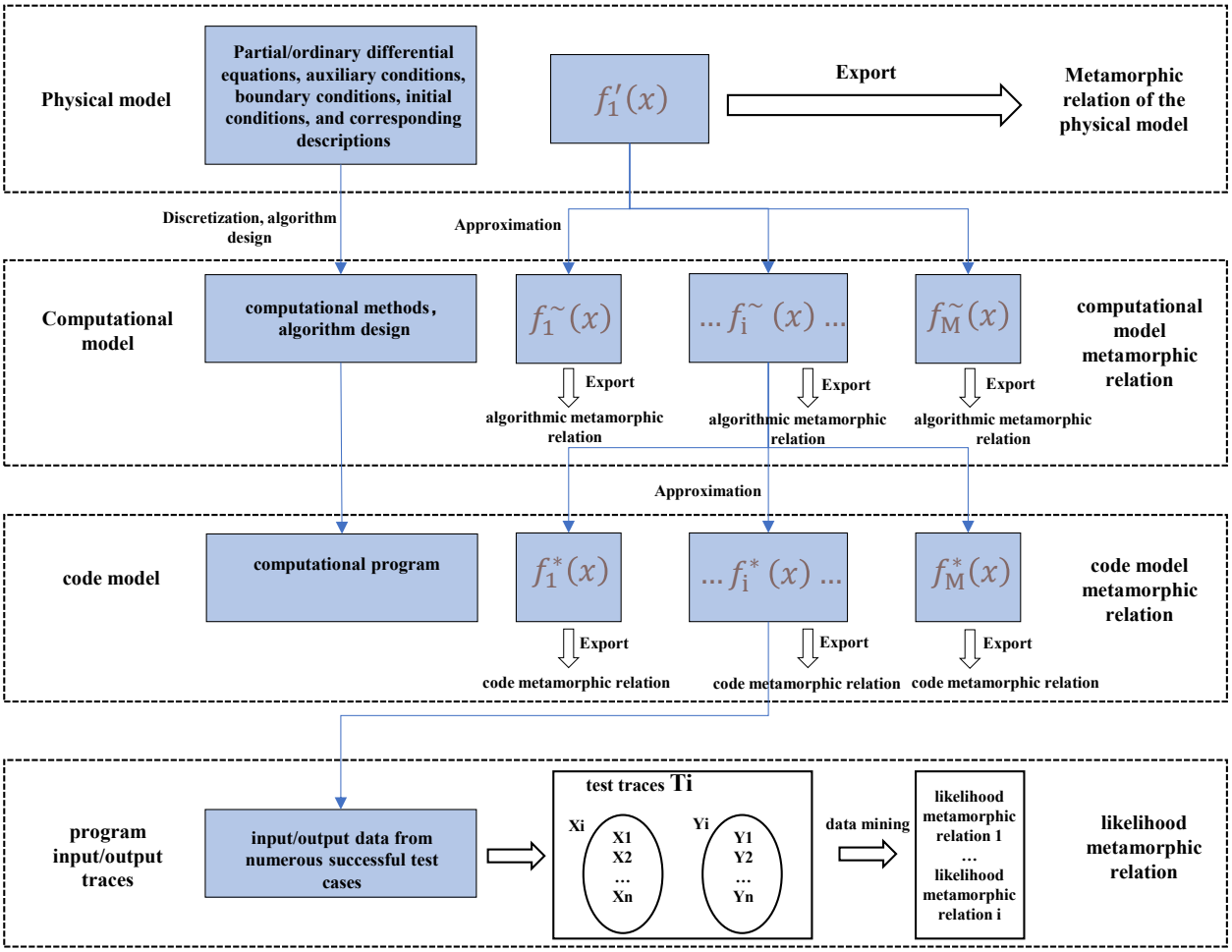


Fig. 1: Hierarchical Classification Model of Metamorphic Relations

and Bieman [14] developed an approach based on control flow graph features, applying machine learning methods for automatic MR prediction. They further improved identification performance with two graph kernel-based feature extension methods [15]. Hardin et al. [34] applied semi-supervised learning for MR detection, utilizing unlabeled data to enhance model classification performance.

B. LLM-Based Identification Methods

Recent studies have begun to explore the application of large language models (LLMs) in automatic MR identification. Tsigkanos et al. [21] were among the first to leverage LLMs to extract MT-related variables from software user manuals, laying the foundation for variable discovery. Zhang et al. [22] achieved automated MR identification using ChatGPT, establishing a basis for intelligent MR recognition. However, their method depends on system descriptions, and the generated results tend to be repetitive and have limited effectiveness, showing sensitivity to prompts and lacking generalizability and automation. In this paper, we improve the quality and adaptability of MR generation through deduplication and staged prompting. Shin et al. [21] were the first to apply LLMs for the automatic identification of executable MRs, advancing the automation of MR recognition. However, their approach relies heavily on high-quality manuals and does not investigate the impact of manual structure on identification performance, resulting in limited generalizability. This work systematically analyzes the sentence structures in manuals, identifies content that aids LLMs in recognizing MRs, and provides optimization suggestions. Luu et al. [24] demonstrated the potential of ChatGPT in MR identification tasks using zero-shot prompts, but the identified MRs were relatively basic, lacking diversity and specificity. This study employs few-shot prompting to significantly enhance the diversity and specificity of identified MRs. Jiaming Zhang et al. empirically evaluated the MR identification capabilities of GPT-3.5 and GPT-4 on 37 subject systems, constructed a unified prompt framework, and discovered new MRs, providing data support

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for large-scale evaluations. However, their prompt design was relatively simple and lacked adaptability. In this paper, we further improve the efficiency and quality of MR identification through staged, multi-turn prompting and a self-verification mechanism.

According to the four-level model , the MRs identified by the aforementioned methods can be systematically categorized based on the types of information they rely on: The methods by Tsigkanos, Zhang, Luu, and Chen's METRIC and METRIC+ primarily depend on system descriptions, domain knowledge, or functional rules, with the identified MRs belonging to the physical model layer; AutoMR focuses on transformations and mutations between algorithmic behaviors, with its identified MRs reflecting characteristics of the computational model layer; the approaches by Kanewala, Bieman, and Hardin are based on control flow graphs and structural features, with their MRs relying on code-level semantic relations and thus belonging to the code model layer; MeMo, MR-Scout, as well as Shin and Jiaming Zhang's methods, combine program execution processes and input-output examples, with the identified MRs constraining program runtime behaviors and falling under the execution trace layer.

The approach proposed in this paper focuses on MRs at the physical model layer. The mined MRs are primarily located at the semantic level, capable of transcending superficial implementations and deeply revealing the essential relationships embedded in the physical laws and algorithmic logic of scientific computing programs, thereby better meeting the complex requirements of the physical modeling domain.

In summary, existing LLM-based MR identification methods are more generalizable than traditional approaches and are not restricted by programming language or application domain. However, they still have limitations in terms of the effectiveness of the identified MRs. To address these issues, this paper proposes strategies such as staged prompting, a self-verification mechanism, and manual structure optimization, which effectively improve the efficiency and quality of automatic MR identification.

IV. Research Methodology

This section presents the research hypotheses, methodology, and illustrative examples.

A. Research Hypotheses

This study is based on the following research hypotheses:

- Large language models possess the capability to identify metamorphic relations (MRs) at the physical model level in the thermal-hydraulic safety design program for pebble-bed high-temperature gas-cooled reactors. Previous studies have demonstrated that LLMs can automatically identify MRs from software documentation; this paper further assumes that such capability also applies to the identification of physical model MRs in the thermal-hydraulic safety design program for pebble-bed high-temperature gas-cooled reactors.
- Structured prompts combined with a self-verification mechanism can enhance the effectiveness and innovativeness of MR identification by LLMs. This paper assumes that through staged prompting and a self-verification mechanism, LLMs can be systematically guided to summarize the properties of the thermal-hydraulic safety design program, thereby improving the accuracy, innovativeness, and interpretability of MR identification.
- The software manual for the thermal-hydraulic safety design program for pebble-bed high-temperature gas-cooled reactors contains heuristic information beneficial for MR identification. It is assumed that the syntactic structures and parameter descriptions within the manual provide effective informational support for LLM-based MR identification.

B. Methodology

The workflow of this study is illustrated in Figure 2. The process is centered around large language models (LLMs) and comprises two main stages: metamorphic relation (MR) generation and MR verification. Compared with existing prompting and overall pipelines, our method excels through a self-verification mechanism and a looped optimization workflow with expert participation.

1)Metamorphic Relation Generation Stage:This stage consists of *data preprocessing*, *prompt construction*, *LLM interaction*, and *MR extraction*.

(i) Data Preprocessing:The software manual is first divided into page-based blocks. Control characters and special symbols are filtered out, and pages containing only pagination information without physical variable content are removed. This prevents excessively long inputs that could hinder LLM reasoning efficiency and result quality, while also reducing noise unrelated to MR identification.

(ii) Prompt Construction:To enable the automatic identification of MRs from the software manual, this study constructs a structured prompt framework that guides the LLM to simulate the expert reasoning process, automatically extract key attributes of the SUT, and summarize its metamorphic relations. As shown in Figure 3, the MR identification prompt template consists of the following four components:

- *Role Definition*: The prompt explicitly assigns a specific role to the LLM, effectively guiding its output.
- *Concept Introduction*: The prompt includes introductions to the concepts of metamorphic testing and metamorphic relations, helping the LLM develop a deeper understanding so that the generated MRs are more likely to satisfy their formal definitions.
- *Structured MR Output*:To avoid unclear or poorly readable MR expressions, a structured MR example in a standardized JSON format is included in the prompt. Through few-shot prompting, the model can generate MRs containing input relations, output relations, and relevant properties in the expected format, improving the standardization and comprehensibility of the results. Structured outputs also facilitate subsequent automated

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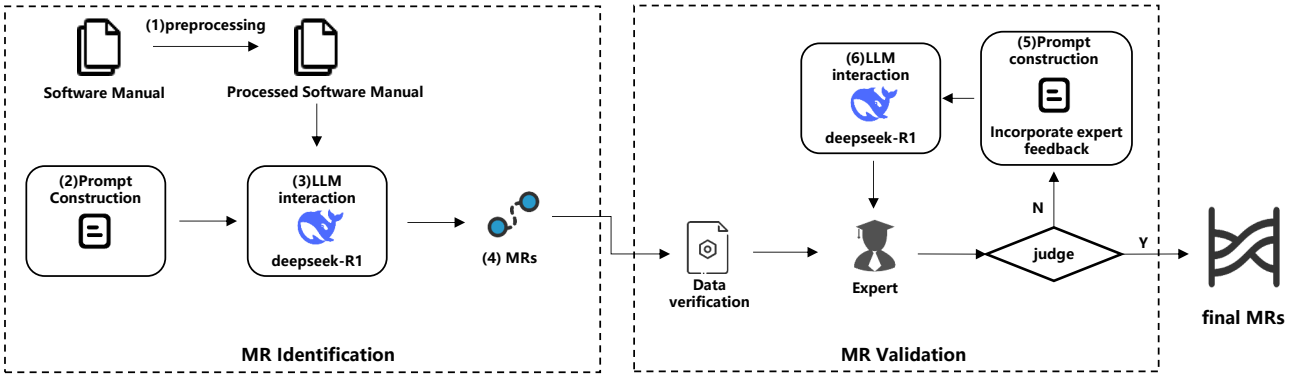


Fig. 2: LLM-Based Workflow

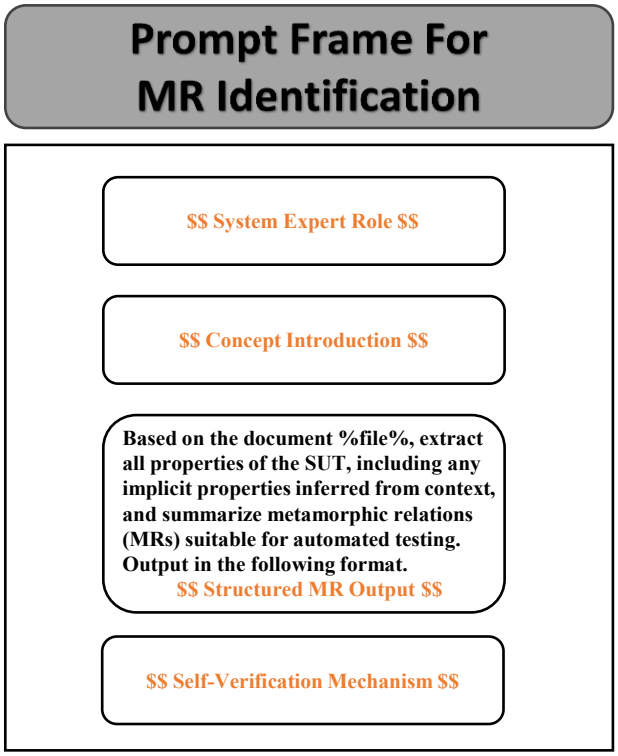


Fig. 3: Prompt Frame For MR Identification

processing, evaluation, and comparative analysis among different MRs. The relevant output format is illustrated in Listing 1.

Listing 1: Example of structured MR output in JSON format

```
{
  "name": "MR01",
  "description": "Input: Increase mass flow rate
      ; Output: Outlet pressure increases",
  "properties": [
    "Mass flow rate",
    "Outlet pressure"
  ]
}
```

- *Self-Verification Mechanism*:This study designs a structured self-verification mechanism (see Figure 4) to guide the LLM in reviewing and examining the extracted system attributes and MRs from multiple perspectives. This mechanism not only helps ensure that the MRs are genuinely derived from the software manual but also explicitly encourages the model to conduct more logical self-checks, thereby improving the rationality, consistency, and interpretability of the output. Specifically, it covers three aspects:
  1) Property Integrity: Whether all key physical properties from the document are covered. The corresponding self-verification question is: "Do the MRs cover the important physical properties in the manual?"
  2) Logical Consistency: Whether the causal relationships and mathematical expressions between inputs and outputs are reasonable. The self-verification question is: "Can the input transformations of the MRs reasonably lead to output changes? Are the mathematical expressions rigorous?"
  3) Domain and Common Sense Consistency: Whether the MRs conform to general laws and common sense in the thermal-hydraulic domain, and whether they depend on reasonable assumptions. The self-verification question is: "Do the MRs comply with basic domain principles? Are additional assumptions required?"

(iii) LLM Interaction:The prompts constructed in the previous subsection are used to guide the large language model in MR identification, as shown in Algorithm 1. The inputs include the target LLM, the interaction termination condition (cond), and the prompts developed earlier. First, a variable history is initialized to record the interaction history. The prompts are input into the LLM in stages: In the first stage, prompts

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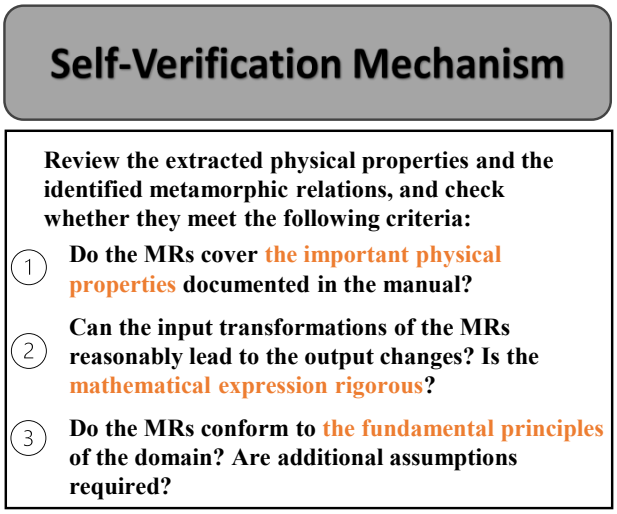


Fig. 4: Self-Verification Mechanism

containing role definition, concept introduction, and structured MR output are provided to guide the LLM in the preliminary identification of metamorphic relations. In the second stage, prompts related to the self-verification mechanism are input, prompting the LLM to conduct self-examination and validation of the identified MRs. In each interaction round, the current content and history are used as input, the identification result is stored in response, and history is updated accordingly. This process is repeated until the termination condition cond is met (for example, when the required number of MRs is reached).

(iv) MR Extraction:To extract all MRs from the LLM interaction results, a list for storing MRs is first initialized as empty. Next, the interaction results are converted to Markdown-formatted text and parsed into an abstract syntax tree (`md_ast`). Each node in the `md_ast` is traversed; nodes identified as JSON code blocks are judged to contain the MRs generated by the LLM, and their content is extracted and stored in the MR list. This process is repeated until the complete set of MRs is obtained.

2)Metamorphic Relation Verification:This stage aims to verify the completeness and correctness of the identified MRs. The process is as follows: For each generated MR, corresponding test cases are automatically constructed and used for initial verification through program testing. Subsequently, nuclear domain experts are organized to conduct a manual review of the MRs. If an MR passes expert review, it is included in the final results; if not, the prompts are revised according to expert feedback and re-input to the LLM to initiate a new round of interaction. To prevent infinite correction loops, the process is limited to a maximum of three iterations. After three rounds of review, the resulting set of MRs is taken as the final output.

The prompts used in this stage are defined as MR verification prompts, as shown in Figure 5. Their structure includes four components: role definition, expert feedback, structured

Algorithm 1: Interactive Extraction of Metamorphic Relations Using LLM

Input: Target LLM llm, Termination Condition $cond$, Prompts set $\mathcal{P}_1$ (identification), $\mathcal{P}_2$ (self-verification)
Output: MR generated by LLM MR

Function `interaction`(llm, $cond$, $\mathcal{P}_1$, $\mathcal{P}_2$)**:**
  history ← empty_list();
  while *not* $cond$ **do**
    $content \leftarrow \mathcal{P}_1$; // ⟶ **Stage 1: Metamorphic Relation Identification**
    $response \leftarrow llm$.interact($content, history$);
    $history$.update($content, response$);
    $content \leftarrow \mathcal{P}_2$; // ⟶ **Stage 2: Self-Verification**
    $response \leftarrow llm$.interact($content, history$);
    $history$.update($content, response$);
  end
  return history;

Function `extract_MR`($interact_res$)**:**
  $MR \leftarrow$ empty_list();
  $md_ast \leftarrow$ parse($interact_res$);
  foreach $node$ *in* md_ast **do**
    if $node$ *is json block* **then**
      $MR \leftarrow$ extract_json($node$);
    end
  end
  return MR;

$output \leftarrow$ `interaction`(llm, $cond$, $\mathcal{P}_1$, $\mathcal{P}_2$);
$MR \leftarrow$ `extract_MR`($output$);
return MR;

MR output, and the self-verification mechanism. Compared to the prompts used in the MR identification stage, the verification prompts additionally incorporate expert feedback, and the self-verification mechanism is adjusted to directly address expert comments, more effectively guiding the LLM to revise and improve the MRs.

The content of the self-verification mechanism is also adapted, primarily to prompt the LLM to further examine whether the revised MRs fully meet the requirements raised in the expert feedback, both in form and definition.

C. Example

The following example illustrates the workflow of the proposed method using a thermal-hydraulic design program. First, the preprocessed software manual and the designed prompts are input into the LLM, which outputs a series of metamorphic relations (MRs) in JSON format. Subsequently, the self-verification mechanism is used to guide the LLM in reviewing these MRs, with the revised results marked as [Revised MR]. The generated MRs are then manually deduplicated (as shown in Figure 6). For each identified

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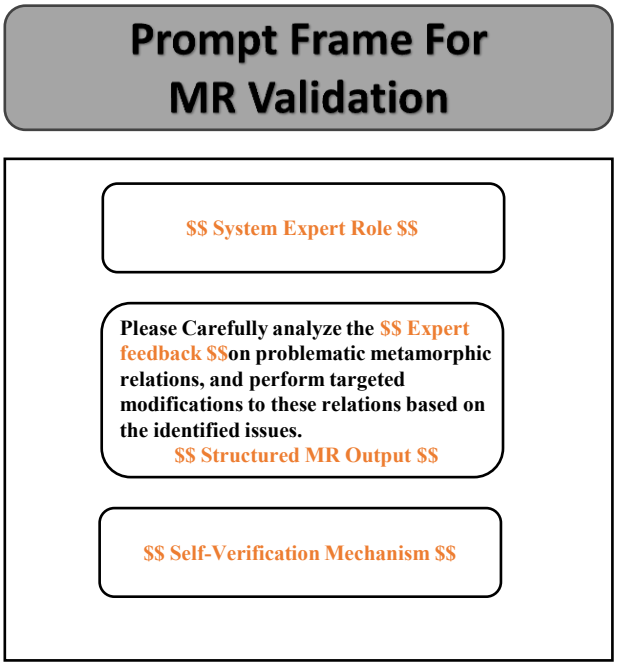


Fig. 5: Prompt Frame For MR Validation

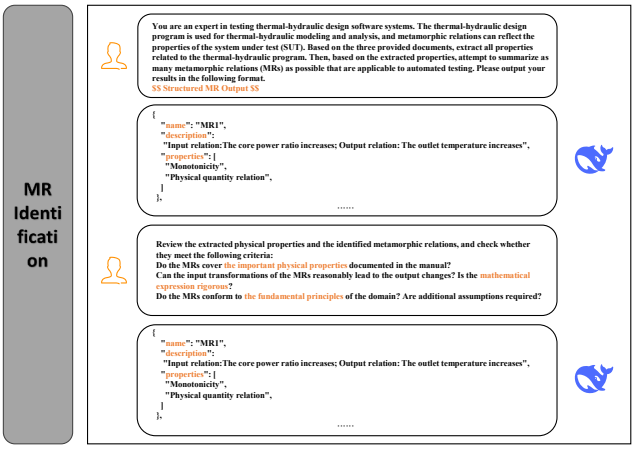


Fig. 6: MR Identification

MR, corresponding test cases are automatically generated and preliminarily validated through program execution. Regardless of the validation results, all MRs must be submitted for domain expert review. Based on expert feedback, the prompts are revised and input into the LLM again to further refine the identification results.

Figure 7 demonstrates the revision process for an MR identified by experts as problematic. MR20 was intended to reflect the relationship between dynamic viscosity and primary side flow rate. However, experts pointed out that dynamic viscosity cannot be directly modified via program

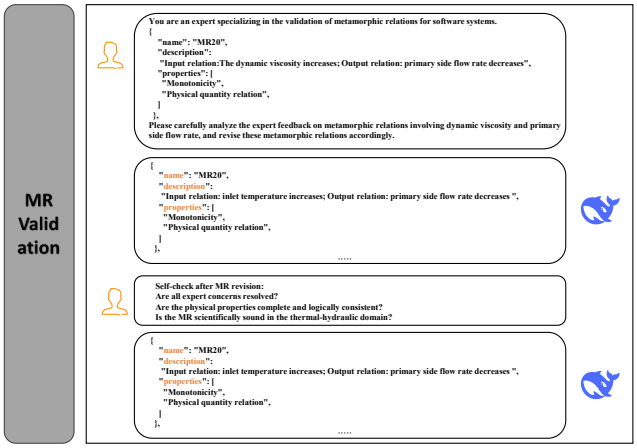


Fig. 7: MR Validation

parameters. Therefore, the expert feedback and this MR are input into the LLM as prompts. Through further analysis of the software manual, the LLM identifies that changes in dynamic viscosity are actually caused by variations in inlet temperature, and accordingly revises the MR to represent the relationship between inlet temperature and primary side flow rate.

V. Experimental Design

Domain experts identified a total of 44 metamorphic relations (MRs) for the thermal-hydraulic safety design program of pebble-bed high-temperature gas-cooled reactors using manual mining methods. In this study, these 44 MRs serve as the reference set for experimentation.

A. Research Questions

The experimental study aims to assess the capability of large language models in identifying MRs for the thermal-hydraulic safety design program of pebble-bed high-temperature gas-cooled reactors. The investigation is structured around four main research questions:

RQ1: How effective is the proposed method in identifying known MRs and discovering new MRs?

To systematically evaluate the effectiveness of the proposed method, this paper not only compares the MRs automatically identified by the LLM with those manually mined by experts, but also systematically benchmarks against current mainstream LLM-driven MR identification approaches. The evaluation consists of the following aspects:

First, the method's performance in recognizing known MRs is evaluated by examining coverage, omissions, and their causes. Second, its capability to discover new, practically valuable MRs beyond traditional approaches is assessed in terms of rationality, completeness, and verifiability.

RQ2: Can the proposed method significantly improve MR identification performance compared with existing methods?

To address this research question, the proposed method is systematically compared with existing approaches, with a par-

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ticular focus on differences in the accuracy and completeness of MR identification results.

RQ3: How do different components of the prompt framework affect the LLM's MR identification performance?

To validate the rationality of the prompt design, ablation experiments are conducted to analyze the specific impact of each component in the prompt framework on the LLM's MR identification performance. coverage, accuracy, and innovation rate are used as evaluation metrics to systematically assess the role of each component in enhancing identification capability and fostering novel discoveries, thus clarifying the contribution and impact of each component on overall performance.

RQ4: Which syntactic structures in the software manual contribute most to MR identification?

To answer this question, the study analyzes the impact of different sentence structures in the software manual on the LLM's ability to identify MRs, aiming to pinpoint effective structures that facilitate relation extraction. The findings provide a foundation for subsequent document selection and standardized writing, while also improving identification efficiency in limited-token scenarios.

B. Experimental Design

1)Experimental Subjects:To validate the proposed method, we used manuals from three subprograms of the thermal-hydraulic safety analysis system for pebble-bed high-temperature gas-cooled reactors (TABLE I). This system addresses key issues such as core thermal behavior, coolant flow, accident safety assessment, and design optimization, and includes: (i) the *Multi Program*—for dynamic simulation and prediction of thermal-hydraulic parameters and core temperature fields at multiple scales; (ii) the *Thermal Design Program*—for modeling and simulating key equipment in the primary and secondary circuits; and (iii) the *Accident Program*—for comprehensive multi-physics simulation and evaluation of main-loop behavior under accident scenarios.

TABLE I: Experimental Subjects

Project	Version	Manual	Total Pages	LOC
Multi	1.1.2	3	75	39804
Thermal	1.0.2	3	69	44879
Accident	1.0.6	3	73	66868

2)Comparison Method:To verify the effectiveness and advancement of the proposed approach, the MR identification method proposed by Jiaming Zhang et al. is selected as the baseline for comparison in this experiment. This method validates the MR identification capability of LLMs on a large scale by employing a unified prompt framework. In this study, it is used as a benchmark to systematically compare MR identification results for the thermal-hydraulic safety design program of pebble-bed high-temperature gas-cooled reactors, thereby highlighting the advantages of the proposed method in terms of coverage, accuracy, and innovation.

3)Evaluation Metrics:Four evaluation metrics are adopted in this study: *coverage rate(CR)*, *accuracy rate(AR)*, *innovation rate(IR)*, and *F1-score*.*CR* refers to the proportion of LLM-identified MRs that are consistent with those identified by domain experts. Although the concept of metamorphic relations is introduced to the LLM through prompts, it may still generate relations that do not match actual or valid MRs.*CR* is calculated as Equation (1):

$$CR = \frac{|\mathrm{MR}_{our} \cap \mathrm{MR}_{sp}|}{|\mathrm{MR}_{sp}|} \times 100\% \tag{1}$$

where $\mathrm{MR}our$ is the number of metamorphic relations identified by the LLM, and $\mathrm{MR}sp$ is the number of metamorphic relations identified by domain experts.

AR refers to the proportion of metamorphic relations identified by the LLM that correctly reflect the properties of the system under test. *AR* is calculated as Equation (2):

$$AR = \frac{|\mathrm{MR}_{our} \cap \mathrm{MR}_{True}|}{|\mathrm{MR}_{our}|} \times 100\% \tag{2}$$

where MR_{True} is the set of correct metamorphic relations.

IR measures the proportion of newly discovered metamorphic relations among all MRs identified by the LLM. In this study, such MRs are referred to as New MRs. New MRs help to reveal previously undiscovered complex characteristics in the thermal-hydraulic safety design program for high-temperature gas-cooled reactors, thereby enhancing the adequacy of testing. At the same time, New MRs can complement existing MRs, reducing manual effort while improving the effectiveness of MR identification.*IR* is calculated as Equation (3):

$$IR = \frac{|\mathrm{NMRs}|}{|\mathrm{MR}_{our}|} \times 100\% \tag{3}$$

$NMRs$ refers to the newly discovered metamorphic relations.

F1-score is used to comprehensively evaluate the performance of the method in the MR identification task. *F1-score* is calculated as Equation (4):

$$F1_{score} = 2 \times \frac{CR \times AR}{CR + AR} \times 100\% \tag{4}$$

AR stands for accuracy rate, and *CR* stands for coverage rate.

4)Experimental Parameter Settings:In this study, DeepSeek-R1 is selected as the base model and accessed via API. Based on preliminary experimental results and relevant literature, the number of repeated experiments for each target program is set to 30, balancing output diversity and experimental efficiency while reducing the randomness and contingency of the model's output. For each program, all MRs generated across experimental rounds are deduplicated, and the final number of MRs is counted when no new relations are generated after 30 experiments, ensuring the comprehensiveness and stability of the results.

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VI. Experimental Results and Discussion

A. RQ1: How effective is the proposed method in identifying known MRs and discovering new MRs?

To evaluate the effectiveness of the proposed method, experiments were conducted on the Multi, Thermal, and Accident programs, and the identified MRs were compared with the expert benchmark set. As shown in Table 2, all known MRs were fully covered, and each program produced eight new, non-redundant MRs independently discovered through the self-verification mechanism. Compared with existing methods, the proposed approach achieved broader coverage and identified novel MRs of distinct types, demonstrating its comprehensiveness and strong capability for new MR discovery.

TABLE II: Comparison of MRs identified by experts, existing methods, and our method in each program

	Expert	Existing Method		SVPrompt-MR	
Program	Identified	Covered	Novel	Covered	Novel
Multi	14	10	5	14	8
Thermal	12	8	3	12	8
Accident	18	16	4	18	8

Furthermore, Figure 8 visually illustrates the consistency between the identified MRs and the benchmark set, indicating that the proposed method achieves comprehensive recognition of known MRs across all programs. Figure 9 compares the new MR discovery capabilities of different methods. In terms of innovation rate, the proposed method achieves 36.36% for Multi, 40.0% for Thermal, and 30.76% for Accident, all significantly higher than those of existing methods, thereby validating its advantage in identifying new relations.

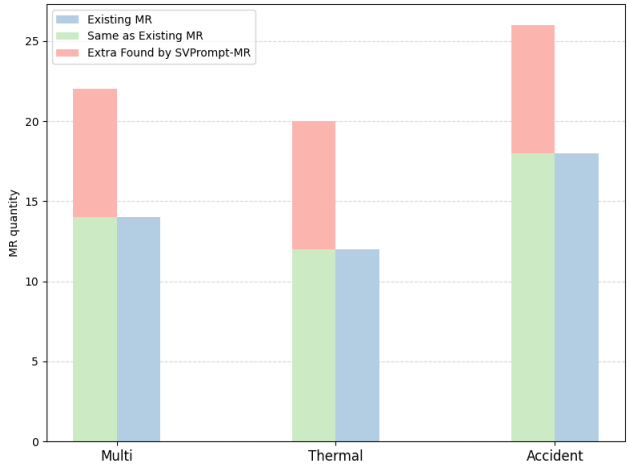


Fig. 8: Completeness performance of metamorphic relations identified by the proposed method

The largest gap in new MR identification occurs in the Accident program (30.76% vs. 20.0%). This stems from the manual's complex structure and implicit logic, which hinder existing methods. In contrast, our self-verification mechanism helps the LLM parse such complexity, enabling discovery of more novel MRs.

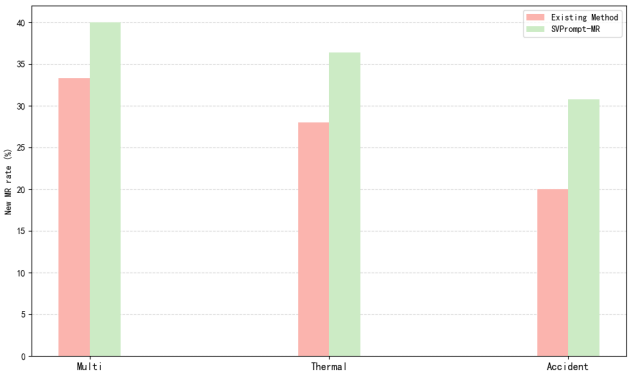


Fig. 9: Proportion of newly identified metamorphic relations in the three programs

Answer to RQ1: The proposed method not only achieves complete identification of existing metamorphic relations but also successfully uncovers multiple potentially valuable new MRs, demonstrating strong capabilities in both coverage and novel discovery.

B. RQ2: Can the proposed method significantly improve MR identification performance compared with existing methods?

To comprehensively evaluate the performance improvements achieved by the proposed method in the MR identification task, a systematic comparison with existing methods was conducted, focusing on key metrics such as *coverage*, *accuracy*, and *F1-score* across the Multi, Thermal, and Accident programs. The detailed results are shown in Table 3.

As indicated in Table 3, the proposed method significantly outperforms existing methods across all evaluation metrics. In terms of *coverage*, the proposed method achieved 100% for all three programs, fully covering the expert-identified MRs, whereas the existing method achieved only 57.14%, 83.33%, and 88.88%, with varying degrees of omission. For *accuracy*, the proposed method attained 81.81%, 90.00%, and 76.92% in Multi, Thermal, and Accident, respectively, which are markedly higher than the existing method's 45.45%, 60.00%, and 69.23%. This demonstrates that the proposed approach can effectively reduce the generation of invalid MRs and improve the reliability of the identification results. Regarding the *F1-score*, the proposed method achieved 89.99%, 94.73%, and 86.95%, representing a substantial improvement compared to the existing method's 50.63%, 69.77%, and 77.83%, and indicating a significant overall advantage in performance.

The main reasons for this significant improvement are: the staged prompts and structured examples better guide the LLM to mine comprehensive and effective MR information from the documentation, while the self-verification mechanism enhances the model's reasoning and reflection capabilities,

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TABLE III: Performance comparison of our method and the existing method on three programs

Method	Program	Coverage Rate (%)	accuracy Rate (%)	F1-score (%)
Existing Method	Multi	57.14	45.45	50.63
	Thermal	83.33	60.00	69.77
	Accident	88.88	69.23	77.83
SVPrompt-MR	Multi	100.00	81.81	89.99
	Thermal	100.00	90.00	94.73
	Accident	100.00	76.92	86.95

enabling multi-perspective self-checks and improving the accuracy of MR identification results.

Answer to RQ2:The proposed method demonstrates significantly superior performance compared to existing approaches in the MR identification task, exhibiting stronger accuracy and effectiveness in MR recognition.

C. RQ3: How do different components of the prompt framework affect the LLM's MR identification performance?

To analyze the impact of prompt design on MR identification, ablation experiments were conducted on the components in both MR identification and verification stages. In the identification stage, *Prompt-A*, *Prompt-B*, *Prompt-C*, and *Prompt-D* represent prompts with removed role definition, concept introduction, structured MR output, and self-verification mechanism, respectively. As shown in Figure 10, the full prompt, which includes all components, achieves the best performance in coverage, accuracy, and innovation rate. The self-verification mechanism is critical for enhancing MR recognition, with its absence leading to the largest decrease in innovation rate. Removing concept introduction notably reduces coverage, indicating its role in improving domain understanding. Structured output improves both accuracy and innovation rate by standardizing the output, while role definition, though having a smaller impact, still reduces all metrics, suggesting its foundational importance in guiding task focus.

Ablation experiments (Fig. 11) show that the full prompt achieves the highest accuracy and innovation, confirming the necessity of all components. Removing expert feedback leads to the largest drop, highlighting its value for domain reasoning; excluding self-verification also degrades logical consistency. Structured output removal moderately affects standardization, while omitting role definition has minimal impact. Coverage remains 100% since no new MRs are generated in this stage.

The final MRs result from the combined effects of both identification and verification stages. Prompt ablation in verification has an "amplification effect," where removing expert feedback produces more errors and reduces innovation. This trend mirrors the identification stage results, showing that well-designed prompts in both stages are essential for maximizing LLM performance in complex MR identification and verification.

Answer to RQ3:Each part of the prompt framework is vital for LLM-based MR identification, with self-verification and expert feedback playing key roles in enhancing accuracy and innovation. High-quality MR identification and verification require well-designed and complete prompts in both stages.

D. RQ4: Which syntactic structures in the software manual contribute most to MR identification?

This study systematically analyzes the impact of different content and sentence structures in the software manual on LLM-based MR identification. Using chapter-wise ablation and fine-grained categorization, the manual is decomposed into definitional, procedural, constraint, causal statements, mathematical formulas, and triples. By comparing changes in MR identification metrics (coverage, accuracy, and innovation) across ablation scenarios, key chapters and sentence types contributing most to MR identification are identified, offering guidance for document selection and standardized writing.

As shown in Figure 12, different sentence structures exhibit significant differences in MR identification metrics. Causal statements achieve the best accuracy and coverage, highlighting their guiding role in enabling LLMs to extract high-quality MRs. Mathematical formulas provide clear definitions of variables and physical relationships, laying the foundation for identifying complex MRs. The triple structure demonstrates distinct advantages in the identification of new MRs, especially in scenarios involving complex variable relationships. In contrast, definitional and constraint statements provide limited support for the identification of complex MRs, while procedural statements mainly assist in the recognition of operation-related MRs.

Further analysis indicates that expressions such as "if...then..." or "as...changes..." significantly improve MR identification accuracy and coverage, while mathematical formulas describing physical properties provide direct and verifiable criteria for MRs. Moreover, transforming natural language into triple structures further enhances the model's reasoning ability and innovativeness. These sentence structures not only facilitate the automatic identification of MRs by LLMs but also provide clear guidance for the standardized writing and structural optimization of software manuals.

Answer to RQ4:Sentence structures that express causal relationships, parameter dependencies, or conditional constraints (such as "if...then..." or "as...changes..."), as well as mathematical formulas describing physical properties and triple structures, are the most beneficial for LLM-based MR identification.

E. Cost Analysis and Threats to Validity

The main costs of this study include API usage fees of the large language model and manual expert feedback. All experiments were conducted through the DeepSeek-R1 API, where charges are primarily based on token usage. Each call consumed approximately 0.5K tokens, resulting in a total of about 4.8 million tokens across the three experimental programs. In terms of human cost, the MR verification stage required expert review and revision, with an average of 10

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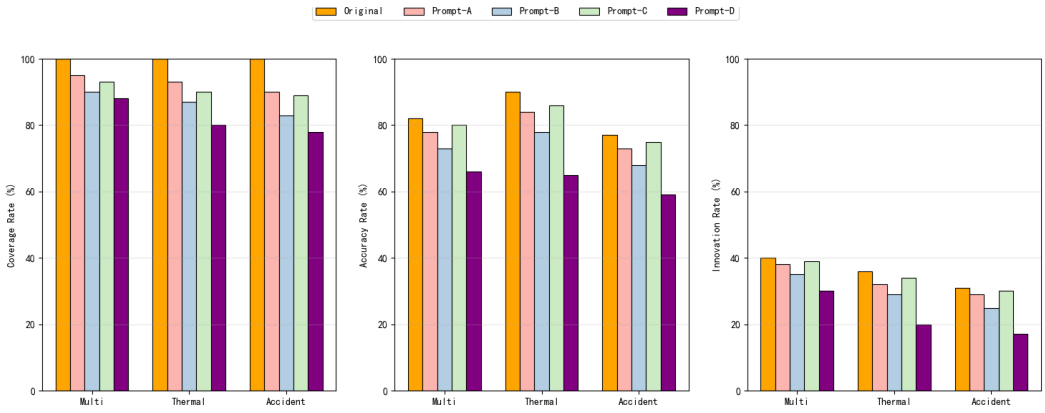


Fig. 10: Ablation experiment results of MR identification prompts (from left to right: CR, AR, IR)

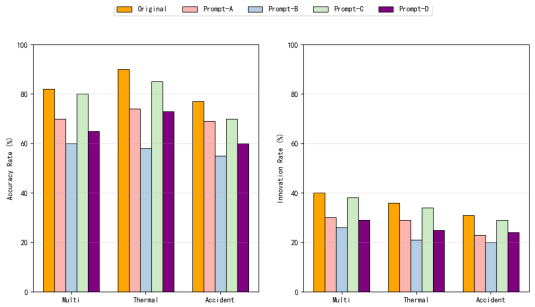


Fig. 11: Ablation experiment results of MR verification prompts (from left to right: AR, IR)

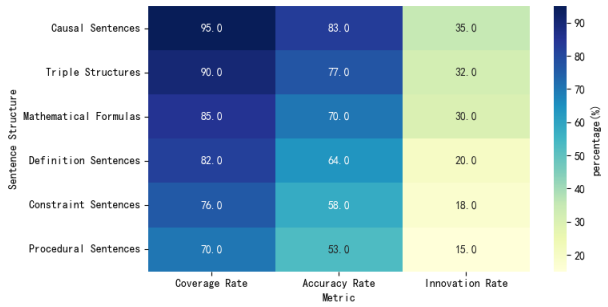


Fig. 12: Effect of different sentence structures on LLM-based MR identification metrics

minutes per MR, involving two experts per round and three rounds of validation.

Although this study focuses on the thermal-hydraulic safety design program, nuclear software, being universally applicable, exhibits characteristics of both numerical and non-numerical software. As such, our method is inherently general. The only domain-specific aspects in the prompt design are the concept introduction and self-verification mechanism, which can be adapted for different fields. For example, in other scientific computing programs, only the domain-specific prompts need adjustment, while the rest of the process remains unchanged. Therefore, the method is highly versatile and can be applied to a wide range of domains. Future research can further test its cross-domain applicability by tailoring the prompts for specific fields and evaluating its effectiveness across different areas.

VII. Conclusion and Future Work

Although the results are promising, this study has several limitations. Validation was limited to the thermal-hydraulic safety design program, so broader cross-domain evaluation is needed to further confirm generalizability. The verification stage still relies on expert review, which restricts scalability despite the effectiveness of the prompt framework. In addition, multiple LLM generation and self-checking rounds increase time and computational costs.

Future work will focus on enhancing scalability and automation. One direction is to introduce an auxiliary LLM as an independent reviewer to pre-screen or evaluate MRs before expert validation, forming a multi-agent collaborative workflow that reduces manual intervention. Moreover, fine-tuning LLMs with knowledge graphs or expert-labeled data could improve domain comprehension and lessen dependence on human experts. Integrating automated MR scoring and verification mechanisms will further enhance efficiency and promote wider applicability across scientific and engineering domains.

VIII. Acknowledgment

This work was supported by the National Natural Science Foundation of China (Grant No. 12575176); the Hunan ProvincialNatural Science Foundation (Regional Joint Program)(Grant No. 2025J170193);and the Hunan Provincial Teaching ReformProject (Grant No.202502000728).

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